ModelIt!

Student Activity Pack

# What's the Weather Like Today?

Watching the Sky and Tracking Weather

Kindergarten | NGSS K-ESS2-1 | Patterns

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

© 2026 Discovery Collective. All rights reserved.

# Table of Contents

ModelIt!


1 Pre-Assessment ..... 3

2 Scoring Rubric ..... 4

3 Component Analysis ..... 5

4 Model Recording Page ..... 6

5 Simulation Observations ..... 7

6 Research & Extend ..... 8

7 Reflection Questions ..... 9

8 STEM Challenge ..... 10

# 1 Pre-Assessment

ModelIt!

**What Do You Know?** Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

**1 Circle today's weather: (picture of sun / picture of clouds / picture of rain / picture of snow)**

**2 Draw what the sky looks like today.**

# 2 Scoring Rubric

ModelIt!

Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

| Skill | Expert (4) | Proficient (3) | Developing (2) | Beginning (1) |
|---|---|---|---|---|
| **Identifying Components** | EXPERT<br>Correctly identifies and sorts ALL components as internal or external with detailed reasoning | PROFICIENT<br>Correctly identifies and sorts MOST components with some reasoning | DEVELOPING<br>Identifies components but has difficulty sorting or explaining | BEGINNING<br>Needs significant help identifying components |
| **Establishing Relationships** | EXPERT<br>Accurately describes ALL positive and negative relationships with clear evidence | PROFICIENT<br>Accurately describes MOST relationships with some evidence | DEVELOPING<br>Identifies some relationships but struggles with direction | BEGINNING<br>Needs help understanding how components connect |
| **Running Simulations** | EXPERT<br>Makes accurate predictions AND explains surprising results using evidence | PROFICIENT<br>Makes reasonable predictions and records clear observations | DEVELOPING<br>Runs simulations but predictions or observations are incomplete | BEGINNING<br>Needs help running simulations or recording data |
| **Explaining Results** | EXPERT<br>Uses multiple pieces of evidence to explain HOW and WHY systems interact | PROFICIENT<br>Uses evidence to explain system interactions | DEVELOPING<br>Gives partial explanations with limited evidence | BEGINNING<br>Needs help connecting evidence to explanations |

# 3 Component Analysis

ModelIt!

### What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

### Sort These Components

Number of Clouds

Amount of Sunshine

| EXTERNAL | INTERNAL |
| --- | --- |
| | |

**Explain your reasoning: Why did you sort each component the way you did?**

# 4 Model Recording Page

ModelIt!

Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

**? Explain ONE relationship in your model. What happens when one component changes?**

# 5 Simulation Observations

ModelIt!

Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

## 1 Scenario 1: Clear Sky

Number of Clouds set to zero

MY PREDICTION

WHAT I OBSERVED

## 2 Scenario 2: Cloudy Sky

Number of Clouds set to high

MY PREDICTION

WHAT I OBSERVED

# 6 Research & Extend

ModelIt!

Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

## Possible New Components

## My Extension

**A Which new component would you add to the model?**

**B How would it connect to the existing components? (Draw or describe)**

**C What evidence or reasoning supports your connection?**

# 7 Reflection Questions

ModelIt!

### The Big Question

**Why is the weather different on different days?** Use evidence from your simulation to support your answers.

# 8 STEM Challenge: Build a Class Weather Station

ModelIt!

## YOUR ENGINEERING MISSION

Create weather tracking tools and use them to observe and record the weather for one week.

### THE SCENARIO

The school principal needs a weather report every day! Your class has been chosen to be the Official School Weather Watchers. You need to build tools to track the weather and report what you see every day for a week. Can your team be the best weather watchers?

### Engineering Design Thinking

1. HOW WILL YOU SHOW IF IT IS SUNNY, CLOUDY, OR RAINY EACH DAY?
2. WHERE WILL YOU RECORD THE WEATHER SO EVERYONE CAN SEE IT?
3. WHAT SYMBOLS WILL YOU USE FOR EACH TYPE OF WEATHER?

### My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

**! Explain your design using scientific evidence from the simulation:**

### Real Career Connection

Meteorologists study the weather and help people know if it will be sunny, rainy, or snowy. They are the weather reporters on TV! They earn $50,000-$100,000/year.